

Formula Sheet: Calculus and Linear Algebra

Series and sums

$$\sum_{k=0}^n k = \frac{n(n+1)}{2}, \quad \sum_{k=0}^n k^2 = \frac{n(n+1)(2n+1)}{6}, \quad (1)$$

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x} \quad (|x| < 1), \quad e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}. \quad (2)$$

The binomial theorem: $(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$.

Integrals

$$\begin{aligned} \int_a^b x^n dx &= \left[\frac{x^{n+1}}{n+1} \right]_a^b, \quad n \neq -1, \\ \int \frac{dx}{x} &= \ln|x| + C, \\ \int_{-\infty}^{\infty} e^{-x^2} dx &= \sqrt{\pi}, \\ \int_0^{2\pi} \int_0^R r dr d\theta &= \pi R^2. \end{aligned}$$

Integration by parts: $\int u dv = uv - \int v du$.

Piecewise definitions

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0, \end{cases} \quad \text{sgn}(x) = \begin{cases} 1 & x > 0, \\ 0 & x = 0, \\ -1 & x < 0. \end{cases}$$

Limits and derivatives

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}, \quad \frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x).$$

Chain rule: $\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$. Taylor's theorem with remainder:

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^{n+1} \quad \text{for some } \xi \text{ between } a \text{ and } x. \quad (3)$$

Matrices

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$\det A = ad - bc, \quad A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \quad \operatorname{tr} A = a + d.$$

The 3×3 identity and a general matrix:

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad M = \begin{bmatrix} m_{11} & m_{12} & \cdots & m_{1n} \\ m_{21} & m_{22} & \cdots & m_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ m_{m1} & m_{m2} & \cdots & m_{mn} \end{bmatrix}.$$

Eigenvalues satisfy $\det(A - \lambda I) = 0$; for symmetric $A \in \mathbb{R}^{n \times n}$ all $\lambda_i \in \mathbb{R}$ and $A = Q\Lambda Q^T$ with Q orthogonal.

Inequalities

$$\left| \sum_{i=1}^n a_i b_i \right| \leq \left(\sum_{i=1}^n a_i^2 \right)^{1/2} \left(\sum_{i=1}^n b_i^2 \right)^{1/2} \quad (4)$$

$$\frac{a+b}{2} \geq \sqrt{ab} \geq \frac{2ab}{a+b} \quad (a, b > 0) \quad (5)$$

Triangle inequality: $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in \mathbb{R}^n$.

Greek letters and operators

$\alpha, \beta, \gamma, \delta, \epsilon, \theta, \lambda, \mu, \pi, \sigma, \phi, \varphi, \omega; \quad \Gamma, \Delta, \Theta, \Lambda, \Pi, \Sigma, \Phi, \Omega.$

Operators and relations: $\partial, \nabla, \infty, \pm, \times, \circ, \cdot, \leq, \geq, \neq, \approx, \equiv, \subseteq, \cup, \cap, \emptyset, \rightarrow, \mapsto, \Leftrightarrow.$